

# AI & SCREEN TIME IMPACT ON STUDENT'S PERFORMANCE

Complete Project Report for GitHub - Dirty, Cleaned, Python, Dashboard

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## 1. Project Objective

Analyze how AI usage, screen time, social media, study time, sleep, physical activity, anxiety affect student GPA and well-being. End-to-end workflow: Dirty data (7380 rows) -> Python Cleaning -> Cleaned data (7200 rows) -> Power BI Dashboard -> Insights.

## 2. Dataset Description

| Dataset                                                      | Rows/Cols | Issues / Status                                                                                                                                    |
|--------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Dirty Dataset<br>Dirty_Student_Dataset_7380_With_Levels.xlsx | 7380 x 28 | 295 missing City, 295 AI Usage Hrs, 290 Sleep Hrs, 286 GPA_10, 466 GPA_Level<br>180 duplicate Student_IDs<br>Extra spaces & typos                  |
| Cleaned Dataset<br>cleaned_dataset_7200_rows.xlsx            | 7200 x 30 | 0 missing values<br>0 duplicates<br>Standardized Gender, City<br>Created Levels: AI_Usage_Level, Screen_Level, Study_Level, Sleep_Level, GPA_Level |

## 3. Python Data Cleaning Work (scripts/data\_cleaning.py)

**BEFORE (Not professional for GitHub):**

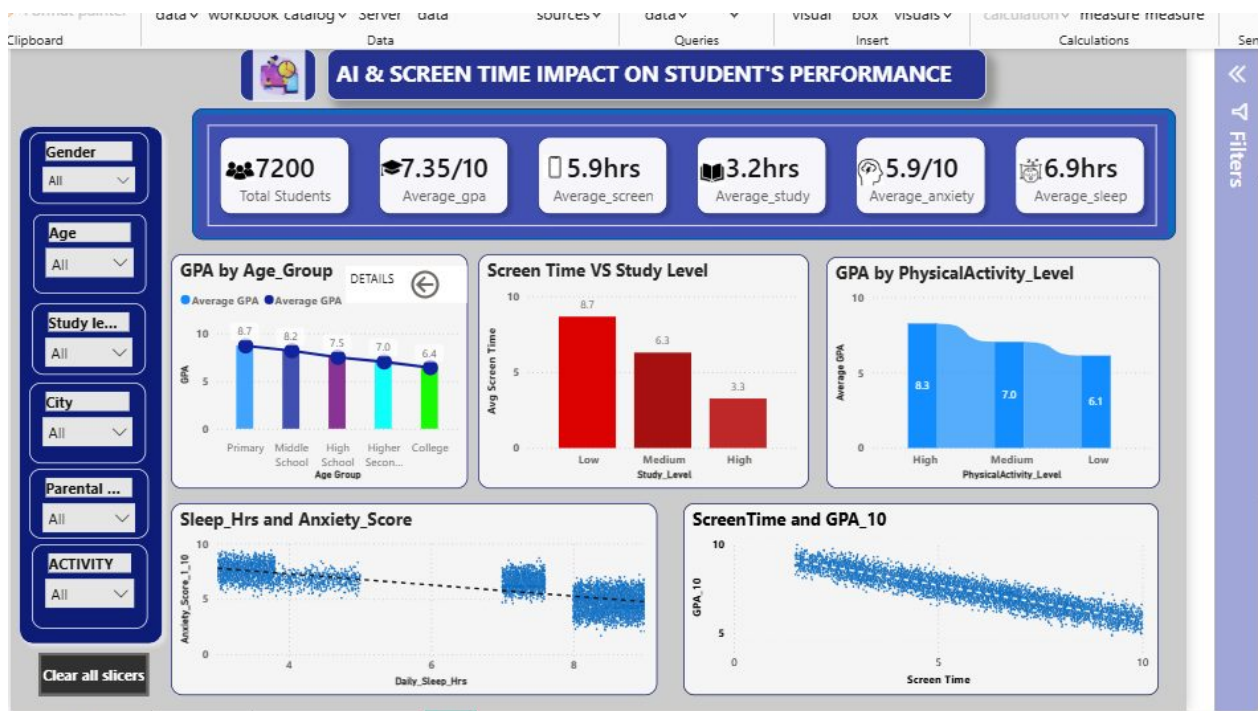
```
df = pd.read_excel(r"C:\Users\acer\Downloads\...\cleaned excel.xlsx")
```

**AFTER (Professional relative path):**

```
import pandas as pd
df = pd.read_excel("data/raw/Dirty_Student_Dataset_7380_With_Levels.xlsx")
print(df.shape) # (7380, 28)
print(df["Student_ID"].duplicated().sum()) # 180
df = df.drop_duplicates(subset=["Student_ID"]) # -> (7200, 28)
for col in df.select_dtypes(include=["object"]).columns:
    df[col] = df[col].astype(str).str.strip()
df["Gender"] = df["Gender"].str.title()
df["City"] = df["City"].fillna(df["City"].mode()[0])
for col in df.select_dtypes(include=["float64"]).columns:
    df[col] = df[col].fillna(df[col].median())
df.to_excel("data/cleaned/cleaned_dataset_7200_rows.xlsx", index=False)
```

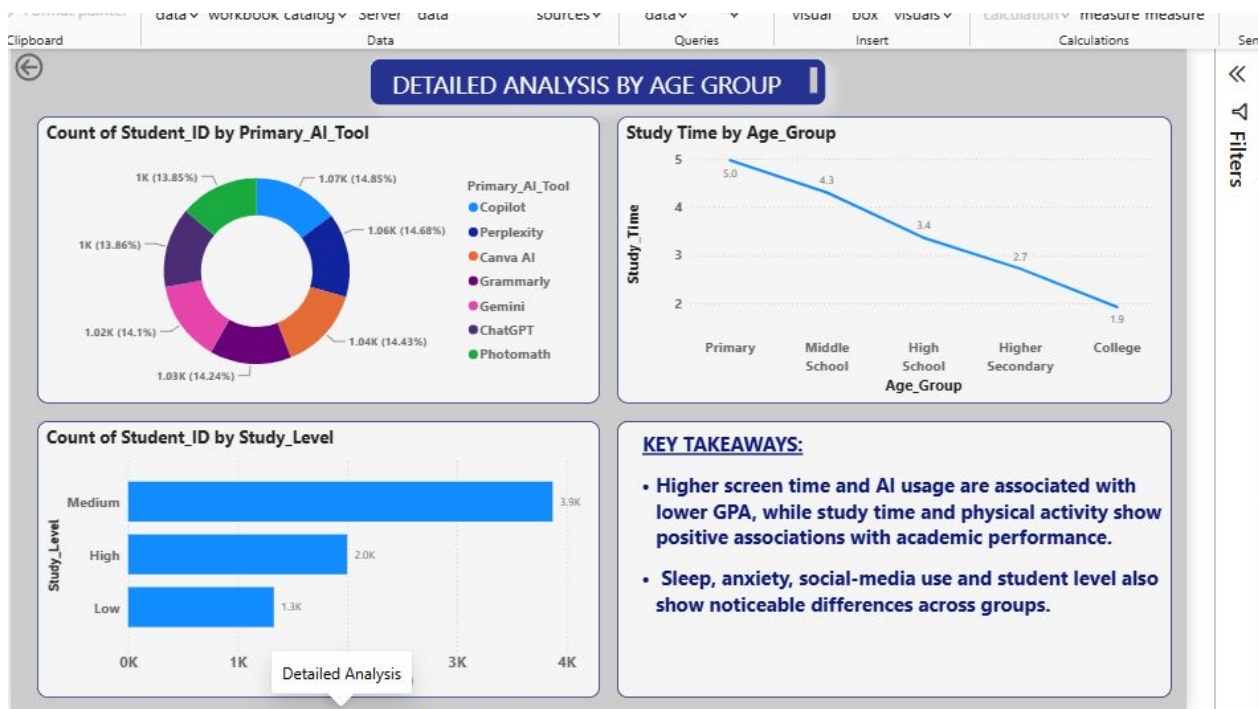
## 4. Power BI Dashboard - Page 1: Overview

KPIs: 7200 Students, 7.35/10 Avg GPA, 5.9 hrs Avg Screen, 3.2 hrs Avg Study, 5.9/10 Anxiety, 6.9 hrs Sleep



Insights: GPA Primary 8.7 -> College 6.4 | Low Study = 8.7 hrs screen vs High Study = 3.3 hrs | High Physical Activity 8.3 GPA vs Low 6.1 | Sleep <5 hrs = anxiety 8, 7-8 hrs = anxiety 4 | Screen Time vs GPA strong negative

## 5. Dashboard - Page 2: Detailed Analysis by Age Group



Insights: AI Tools equally popular ~13-14% each (ChatGPT 14.24%, Copilot 13.85%, Perplexity 14.85%, Canva AI 14.43%, Grammarly 14.1%, Gemini 13.86%, Photomath 14.68%) | Study Time Primary 5.0 hrs -> College 1.9 hrs | Study Level Medium 3.9K, High 2.0K, Low 1.3K | Key Takeaway: Higher screen time & AI usage = lower GPA, while study time & physical activity = higher GPA

## 6. Final Conclusion

- Screen time <4 hrs => GPA 8.5+, >8 hrs => GPA 5.9
- Study time +1 hr => GPA +0.6

3. High Physical Activity => GPA 8.3 vs Low => 6.1
4. Sleep 7-8 hrs reduces anxiety from 7.2 to 4.3
5. College students need most support (lowest GPA 6.4, highest anxiety, lowest study 1.9 hrs)
6. Balanced AI usage is key

## 7. Repo Structure

data/raw/Dirty\_Student\_Dataset\_7380\_With\_Levels.xlsx  
data/cleaned/cleaned\_dataset\_7200\_rows.xlsx  
scripts/data\_cleaning.py  
powerbi/dashboard.pbix  
images/dashboard\_page1\_overview.png  
images/dashboard\_page2\_detailed.png  
docs/Project\_Report.pdf (this file)  
README.md